

Wirtschaftsinformatik - WIN6 - Business Intelligence

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Exercises

For all exercises, unless stated otherwise, model the situation as an *undirected, unweighted, simple graph*.

1. Modeling a collaboration network.

A mid-sized company analyzes weekly collaboration between six units: Sales (S), Marketing (M), Logistics (L), IT (I), Finance (F), and HR (H). The collaboration relations are

$$(S, M), (S, L), (M, I), (L, I), (I, F), (F, H).$$

- (a) Define the node set V and the edge set E .
- (b) Draw the graph by hand.
- (c) Write the adjacency matrix in the order (S, M, L, I, F, H) .
- (d) Determine n , m , the degree of each node, and the average degree.

2. Handshake lemma and feasibility of a degree sequence.

In a B2B platform network with eight firms, the proposed degree sequence is

$$(1, 1, 2, 2, 3, 3, 4, 5).$$

- (a) Use the handshake lemma to decide whether this sequence can belong to an undirected graph.
- (b) If the network has $m = 10$ edges instead, what must the sum of all degrees be?
- (c) Explain in one sentence why every added edge increases the total degree by exactly 2.

3. Shortest paths in a product co-purchase network.

An online retailer observes the following co-purchase network of six products:

$$(P_1, P_2), (P_1, P_3), (P_2, P_3), (P_3, P_4), (P_4, P_5), (P_4, P_6), (P_5, P_6).$$

- (a) Draw the graph.
- (b) Compute all shortest-path distances from P_1 to the other nodes.
- (c) Compute all shortest-path distances from P_4 to the other nodes.
- (d) Determine the diameter of the network.
- (e) Compute the average path length (APL).
- (f) Explain what the APL means in the context of product discovery.

4. Comparing two help-desk escalation structures.

A company compares two possible escalation networks on six agents.

- Network A: a chain

$$1 - 2 - 3 - 4 - 5 - 6$$

- Network B: a star with node 1 in the center

$$(1, 2), (1, 3), (1, 4), (1, 5), (1, 6)$$

- (a) Compute the APL of Network A.
- (b) Compute the APL of Network B.
- (c) Which network is better for fast information diffusion?
- (d) Which network creates a stronger dependency on a single actor?
- (e) Explain the business trade-off between efficiency and robustness.

5. Disconnected network and global efficiency.

A legacy IT landscape has the following communication structure:

$$(A, B), (B, C), (D, E).$$

- (a) Draw the graph.
- (b) Explain why the APL of the whole network is not well-defined.
- (c) Compute the APL of the largest connected component.

- (d) Compute the global efficiency

$$E = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{d(i,j)}$$

using the convention $\frac{1}{\infty} = 0$.

- (e) Interpret the result for integration management.

6. Local clustering coefficient in a customer referral network.

A referral network among six customers is given by

$$(A, B), (A, C), (B, C), (C, D), (D, E), (D, F), (E, F).$$

- (a) Compute the local clustering coefficient of every node.
- (b) Which nodes are most strongly embedded in a local group?
- (c) Which nodes act as potential bridges between two groups?
- (d) Which customer would you approach first for a referral campaign that should spread across the whole network? Justify your answer.

7. Average local clustering vs. global transitivity.

Use the graph from Exercise 6.

- (a) Compute the average local clustering coefficient

$$\bar{C} = \frac{1}{n} \sum_i C_i.$$

- (b) Count the number of triangles in the graph.
- (c) Count the number of connected triples.
- (d) Compute the global transitivity

$$C_{\text{global}} = \frac{3 \cdot \text{\#triangles}}{\text{\#connected triples}}.$$

- (e) Explain why $\bar{C}$ and C_{global} do not have to be identical.
- (f) Discuss one business advantage and one disadvantage of high clustering.

8. The effect of a bridge edge.

Start from the graph in Exercise 6.

- (a) Remove the edge (C, D) . How many connected components does the network now have?
- (b) Describe qualitatively what happens to the APL of the whole network.
- (c) Describe qualitatively what happens to the clustering coefficients of nodes C and D .
- (d) Describe qualitatively what happens to the algebraic connectivity λ_2 .
- (e) Now return to the original graph and add the edge (B, E) instead. Predict the direction of change for APL and clustering.
- (f) Interpret both interventions from the viewpoint of knowledge transfer between departments.

9. Degree distribution in an enterprise application network.

The degrees of ten enterprise software modules are

1, 1, 1, 2, 2, 2, 3, 3, 5, 8.

- (a) Construct the degree distribution $P(k)$.
- (b) Construct the cumulative distribution $P(K \geq k)$.
- (c) Compute the average degree.
- (d) Identify the hub modules.
- (e) Explain one operational risk that arises if a hub module fails.

10. Estimating a power-law exponent.

Assume that the tail of a partner network follows

$$P(k) = Ck^{-\gamma}.$$

Empirically, nodes with degree 2 occur with probability 0.18, and nodes with degree 4 occur with probability 0.045.

- (a) Estimate the exponent γ .
- (b) Rewrite the model in log-log form.
- (c) Explain what the slope means in a log-log plot.
- (d) Interpret what a smaller value of γ would mean for market concentration and systemic risk.

11. Preferential attachment probabilities.

A digital platform currently has five sellers with degrees

$$8, 5, 3, 2, 2.$$

A new seller joins and creates $m = 2$ new links according to preferential attachment.

- (a) Compute the total degree of the current network.
- (b) Compute the attachment probability of each existing seller.
- (c) Which seller is expected to benefit most from the new arrival?
- (d) Explain in words why preferential attachment tends to create hubs over time.

12. When is preferential attachment a good model?

Answer the following conceptual questions.

- (a) Give two Wirtschaftsinformatik contexts in which preferential attachment is a plausible mechanism.
- (b) Give two contexts in which preferential attachment is *not* a plausible mechanism.
- (c) For each non-plausible case, name one alternative mechanism such as homophily, geography, compliance rules, or triadic closure.
- (d) Explain why it is dangerous to assume a power law without checking the underlying business process.

13. Constructing the Laplacian matrix.

Consider an ERP integration network with nodes

$$S, M, I, F$$

and edges

$$(S, M), (S, I), (M, I), (I, F).$$

- (a) Write the adjacency matrix A in the order (S, M, I, F) .
- (b) Write the degree matrix D .
- (c) Compute the Laplacian matrix

$$L = D - A.$$

- (d) Verify that each row sum of L is equal to zero.

14. **Connected components from the Laplacian.**

Consider the matrix

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & -1 & -1 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \end{bmatrix}.$$

- (a) Without drawing the graph first, determine how many connected components the network has.
- (b) Identify which nodes belong to the same component.
- (c) State what this implies for the multiplicity of the eigenvalue 0.
- (d) Give a business interpretation if the nodes represent IT systems inside one company.

15. **The quadratic form of the Laplacian.**

Use the ERP integration graph from Exercise 13. Suppose the vector

$$x = (4, 4, 5, 1)^T$$

describes the digital maturity score of the four systems.

- (a) Compute

$$x^T L x$$

using the formula

$$x^T L x = \sum_{(i,j) \in E} (x_i - x_j)^2.$$

- (b) Explain what a large value of $x^T L x$ means in terms of heterogeneity between connected systems.
- (c) Explain why $x^T L x \geq 0$ must always hold.

16. **Algebraic connectivity.**

Three alternative communication networks on the same set of employees have the following second Laplacian eigenvalues:

$$\lambda_2^{(A)} = 0, \quad \lambda_2^{(B)} = 0.14, \quad \lambda_2^{(C)} = 0.91.$$

- (a) Rank the three networks by overall connectedness.
- (b) Which network is disconnected?

- (c) Which network appears most robust against a small structural disturbance?
- (d) Explain why a small but positive λ_2 indicates a bottleneck.
- (e) Give one managerial interpretation for each case.

17. Interpreting a Fiedler vector.

An IT architecture network has the nodes

(Sales, CRM, Marketing, Middleware, Production, Logistics, Warehouse)

and the approximate Fiedler vector

$$v_2 = (-0.52, -0.47, -0.31, 0.08, 0.43, 0.49, 0.55)^T.$$

- (a) Split the nodes into two groups using the sign of the vector entries.
- (b) Which node appears to be closest to the boundary between the two groups?
- (c) What does this suggest about organizational or technical silos?
- (d) Why is a near-zero entry strategically important?
- (e) Suggest one concrete management action based on this information.

18. Adding an edge using spectral intuition.

Consider the network with nodes

$$A, B, C, D, E, F$$

and edges

$$(A, B), (A, C), (B, C), (C, D), (D, E), (D, F), (E, F).$$

Its approximate Fiedler vector is

$$v_2 \approx (-0.46, -0.46, -0.26, 0.26, 0.46, 0.46)^T.$$

Management can add exactly one new cross-group edge: (A, E) , (C, E) , or (C, F) .

- (a) Which edge would you expect to increase λ_2 the most? Justify your answer using the Fiedler vector.
- (b) Which proposed edge addresses the structural bottleneck most directly?

- (c) Explain why Fiedler-vector-based edge addition can be useful in organizational design.

19. Small-world structure in business networks.

A corporate social network shows *high clustering coefficient* and *low average path length*.

- (a) Explain why this is called a small-world structure.
- (b) Give two advantages of such a structure for innovation diffusion and coordination.
- (c) Give two risks of such a structure for rumor spreading, cyber risk, or systemic dependency.
- (d) Suggest one managerial measure that preserves the benefits while reducing the risks.

20. Integrated case study: knowledge exchange in a consulting firm.

A consulting firm models knowledge exchange between seven consultants by the edge set

$$(A, B), (A, C), (B, C), (C, D), (D, E), (D, F), (E, F), (F, G).$$

- (a) Draw the graph.
- (b) Determine n , m , the degree of every node, and the average degree.
- (c) Compute the local clustering coefficients of nodes C , D , F , and G .
- (d) Identify the node or edge that is most likely to form the main bottleneck.
- (e) Without calculating eigenvectors numerically, explain what sign pattern you would expect from the Fiedler vector.
- (f) Give one concrete recommendation to management if the goal is to reduce silos and improve knowledge transfer.